# Towards Subseasonal-to-Seasonal Drought-to-Flood Prediction

**Lainey Ward**[1], Fiachra O'Loughlin[1], Conor Sweeney[2]

[1]School of Civil Engineering, [2]School of Mathematics and Statistics, University College Dublin

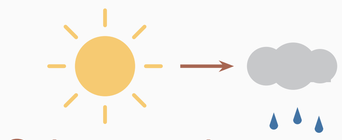
## Introduction

Floods and droughts can occur in sequence — a **drought-to-flood transition (D2F)**. Globally, **~25% of floods are preceded by drought** [2], and socio-economic losses can be up to **8x greater** than from isolated hazards [3]. Forecasting D2Fs at subseasonal-to-seasonal lead times could support early action, but 2 research gaps remain ⟶

No catalogue of D2F events exists for Ireland

↓ Current focus

Subseasonal-to-seasonal forecast skill over Ireland has not been assessed



Subseasonal-to-seasonal D2F prediction

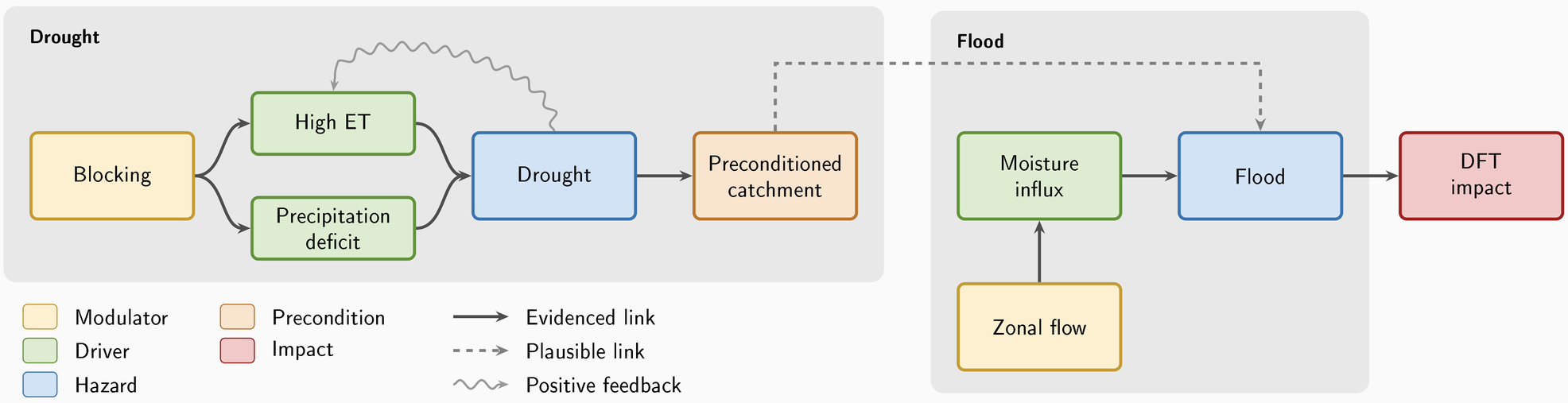



**Fig. 1:** Physical pathway for D2F events, after the framework of Bevacqua et al. [1].

## Methods

We verify 2 m temperature forecasts at **2–6 week** lead times, pooling all reforecast initialisations (2005–2024) per season and lead. Two complementary metrics are used: **Fair CRPSS** (Continuous Ranked Probability Skill Score) [4], scored against ERA5 climatology, and **centred ACC** (Anomaly Correlation Coefficient). Mean bias is removed by leave-one-year-out correction.

$$\mathrm{ACC} = \frac{\sum_{m=1}^{M} w_m \,(f'_m - \bar{f}')(o'_m - \bar{o}')}{\left[\sum_{m=1}^{M} w_m \,(f'_m - \bar{f}')^2 \cdot \sum_{m=1}^{M} w_m \,(o'_m - \bar{o}')^2\right]^{1/2}}$$

$f'_m,\ o'_m$ = forecast/observed anomalies
$\bar{f}',\ \bar{o}'$ = area-weighted spatial means
$w_m = \cos\varphi$ (latitude weight)
$M$ = number of gridpoints

Wilks (2011), Eq. 7.59 [5]

## Data

| | |
|---|---|
| **System** | ECMWF ENS extended-range |
| **IFS cycle** | CY49R1 (2024) |
| **Init. freq.** | Every 2 days |
| **Members** | 11 (1 control + 10 perturbed) |
| **Horizon** | Weeks 2–6 |
| **Variable** | 2 m temperature |
| **Grid** | 0.75° regular lat-lon (MARS) |
| **Reference** | ERA5 + Met Éireann stations |
| **Reforecasts** | 2005–2024 |

## Results

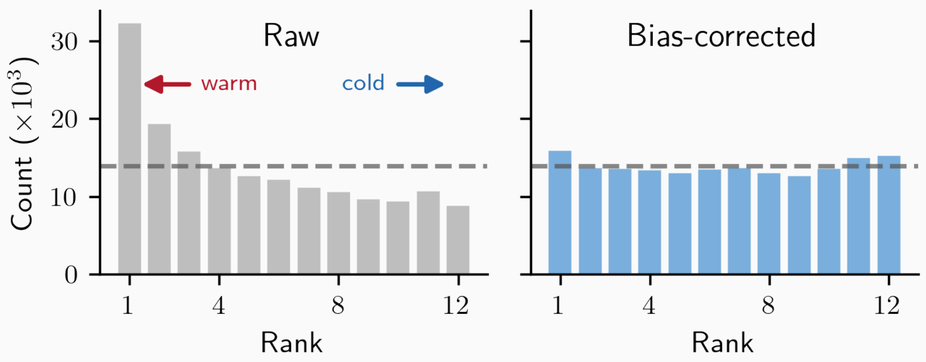



**Fig. 2:** Rank histograms before and after bias correction.

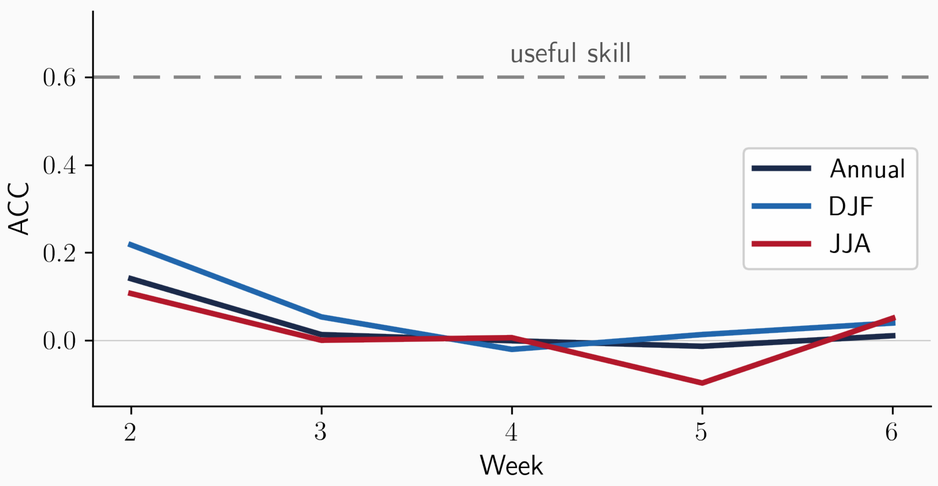



**Fig. 3:** ACC by lead time.

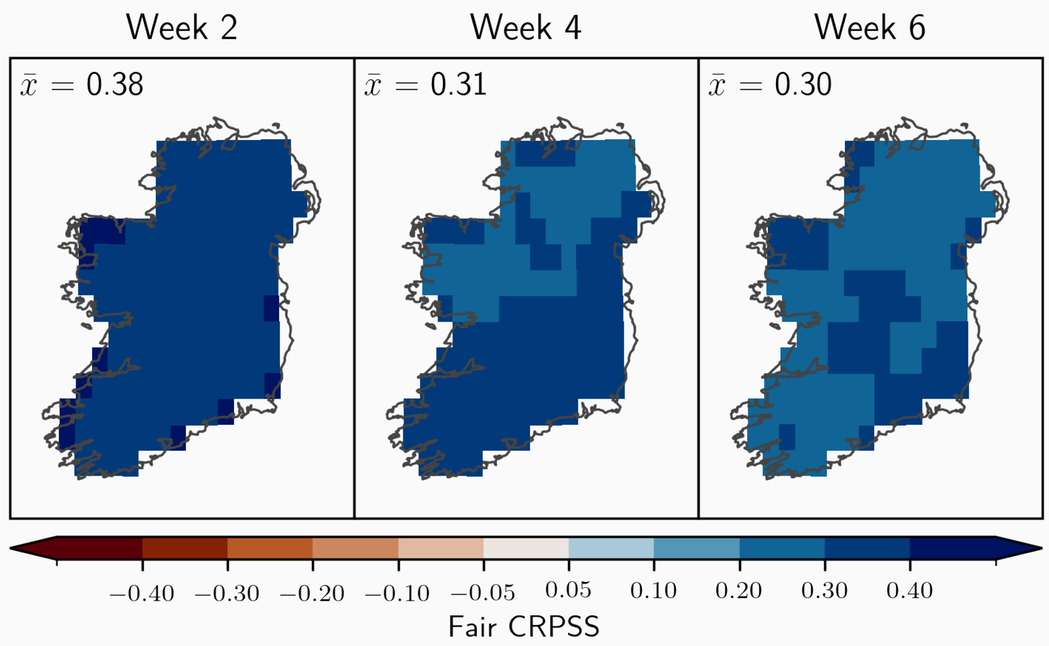



**Fig. 4:** CRPSS by lead time (bias-corrected).

- Your interpretation here
- Your interpretation here
- Your interpretation here

## Future Work

Compare with SEAS5 seasonal forecasts to map skill across the full subseasonal-to-seasonal timescale

Create D2F event catalogue for Ireland; link to catchment and soil properties

⟶ **Identify predictable at-risk catchments**

## References

[1] Bevacqua et al. (2021). *Earth's Future*, 9, e2021EF002340. [2] Matanó et al. (2024). *Environ. Res. Lett.*, 19, 064048. [3] Worou & Messori (2025). *Environ. Res. Lett.*, 20, 104024. [4] Ferro (2014). *Q. J. R. Meteorol. Soc.*, 140, 1917–1923. [5] Wilks (2011). *Statistical Methods in the Atmospheric Sciences*, 3rd ed., Eq. 7.59.